

Course 4541

4 Credits: 4

Program Core

Source-grounded

# Power Electronics

- To introduce students to the basic theory of power semiconductor devices and

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 4541 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 4541.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

## COURSE DASHBOARD

### Official course information

#### Course information

| Item                 | Official information                     |
|----------------------|------------------------------------------|
| Programme            | Diploma in Electrical Engineering        |
| Course code          | 4541                                     |
| Course title         | Power Electronics                        |
| Semester             | 4 Credits: 4                             |
| Course category      | Program Core                             |
| Periods per week     | 4 (L:3 T:1 P:0) Periods per semester: 60 |
| Periods per semester | 60                                       |

| Item         | Official information                                 |
|--------------|------------------------------------------------------|
| Credits      | 4                                                    |
| Prerequisite | See the official syllabus PDF                        |
| Assessment   | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

15 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- To introduce students to the basic theory of power semiconductor devices and passive components, and their practical applications in power electronics.
- To familiarize students with the principle of operation of different power electronic converters and their applications.
- To provide a strong foundation for further study of power electronics

### Course outcomes

| CO  | Official outcome                                                                               | Study level           |
|-----|------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | a particular application. 16 Understanding Summarize the principle and operation of controlled | See official syllabus |
| CO2 | 14 Understanding rectifiers. Summarize the principle and operation of DC and AC Understanding  | See official syllabus |
| CO3 | converters. 12                                                                                 | See official syllabus |
| CO4 | Summarize various inverters and electric drives. 16 Understanding                              | See official syllabus |

### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 2                      |
| CO2 | CO2 2                      |
| CO3 | CO3 2                      |

| Row | Extracted official mapping |
|-----|----------------------------|
| CO4 | CO4 2                      |

## COVERAGE AUDIT

# Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

### Module coverage

| Module   | Official heading                                                | Topic entries | Hours        |
|----------|-----------------------------------------------------------------|---------------|--------------|
| Module 1 | application.                                                    | 4             | As specified |
| Module 2 | Summarize the principle and operation of controlled rectifiers. | 4             | As specified |
| Module 3 | Summarize the principle and operation of DC and AC converters.  | 3             | As specified |
| Module 4 | Summarize various inverters and electric drives                 | 4             | As specified |

## MODULE 1

# application.

This module is presented from the official syllabus scope for Power Electronics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: application.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## devices.

devices. is an official topic in Module 1 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify devices. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, devices. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What devices. means in the official course context                                |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഓട്ടം devices. ഓട്ടംഓട്ടംഓട്ടംഓട്ടം ഓട്ടം  
definition/meaning ഓട്ടംഓട്ടംഓട്ടംഓട്ടം. ഓട്ടംഓട്ടം ഓട്ടംഓട്ടം, steps, application  
ഓട്ടംഓട്ടം ഓട്ടംഓട്ടംഓട്ടം ഓട്ടംഓട്ടം. Technical terms English-ഓട്ടം ഓട്ടംഓട്ടംഓട്ടംഓട്ടം.

## Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off

Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off is an official topic in Module 1 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the operation of scr 3 understanding classify and explain turn on and turn off should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-എം Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off ഓൺ ഓഫ് definition/meaning ഓൺ ഓഫ്. ഓൺ ഓഫ് ഓൺ ഓഫ്, steps, application ഓൺ ഓഫ്. Technical terms English-എം ഓൺ ഓഫ്.

## 4 Understanding methods of SCR.

4 Understanding methods of SCR. is an official topic in Module 1 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding methods of SCR. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding methods of scr. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding methods of SCR. means in the official course context         |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-4 Understanding methods of SCR.

ശ്രീകൃഷ്ണൻ്റെ രചന definition/meaning ശ്രീകൃഷ്ണൻ്റെ രചന. ശ്രീകൃഷ്ണൻ്റെ രചന, steps, application ശ്രീകൃഷ്ണൻ്റെ രചന. Technical terms English-ശ്രീകൃഷ്ണൻ്റെ രചന.

**Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols ( MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)- structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only)**

Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols ( MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods - SCR Turn-On methods -list various methods - basic concepts - gate triggering - forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only) is an official topic in Module 1 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols ( MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic

diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits.

Specifications of SCR- List parameters (Datasheet interpretation only) using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, summarize the idea of datasheets of scr 1 understanding power electronic devices- list various devices -symbols ( mosfet, igbt,gto, lascr, scs, ujt, diac, triac and scr) -mosfet - list different types -structure and working of n-channel enhancement type-igbt-- structure-working principle - applications-ujt- schematic diagram and working- diac & triac-schematic diagram- working- v/i characteristics. silicon controlled rectifier (scr)-structure-v-i characteristics- definitions-holding current- latching current scr turn on and turn off methods -scr turn-on methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-scr turn-off methods- list various methods- basic concepts-thyristor protection circuits-snubber circuits. specifications of scr- list parameters (datasheet interpretation only) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols ( MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods - SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only) means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-  
Understanding Power Electronic Devices- List various devices -Symbols ( MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types - structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC- schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)- structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only)

പ്രതിരോധ ഉപകരണങ്ങളുടെ നിർവ്വചന/അർത്ഥം. പ്രയോഗ ഘട്ടങ്ങൾ, application പദങ്ങൾ. Technical terms English-  
പ്രതിരോധ ഉപകരണങ്ങൾ.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

application.

|                        |                                                           |   |
|------------------------|-----------------------------------------------------------|---|
| M1.01<br>Understanding | Summarize the operation of power electronic devices.      | 8 |
| M1.02<br>Understanding | Explain the operation of SCR                              | 3 |
| M1.03<br>Understanding | Classify and explain turn on and turn off methods of SCR. | 4 |
| M1.04<br>Understanding | Summarize the idea of datasheets of SCR                   | 1 |

Contents:

Power Electronic Devices- List various devices -Symbols ( MOSFET, IGBT, GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics.

Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions- holding current- latching current

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 2

# Summarize the principle and operation of controlled rectifiers.

This module is presented from the official syllabus scope for Power Electronics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Summarize the principle and operation of controlled rectifiers.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.



## 4 Understanding wave controlled rectifier Explain the operation of single phase full

4 Understanding wave controlled rectifier Explain the operation of single phase full is an official topic in Module 2 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding wave controlled rectifier Explain the operation of single phase full using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding wave controlled rectifier explain the operation of single phase full should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding wave controlled rectifier Explain the operation of single phase full means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-4 Understanding wave controlled rectifier

Explain the operation of single phase full bridge rectifier definition/meaning  
operation. Explain the steps, application of the rectifier.  
Technical terms English- Malayalam.

## 5 Understanding wave controlled rectifier Summarize the operation of single phase

5 Understanding wave controlled rectifier Summarize the operation of single phase is an official topic in Module 2 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding wave controlled rectifier Summarize the operation of single phase using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding wave controlled rectifier summarize the operation of single phase should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding wave controlled rectifier Summarize the operation of single phase means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-5 Understanding wave controlled rectifier

Summarize the operation of single phase  $\text{SCR}$  definition/meaning  $\text{SCR}$ .  $\text{SCR}$   $\text{SCR}$   $\text{SCR}$ , steps, application  $\text{SCR}$   $\text{SCR}$ . Technical terms English- $\text{SCR}$   $\text{SCR}$   $\text{SCR}$ .

## 2 Understanding semi controlled rectifiers Outline the operation of three phase

2 Understanding semi controlled rectifiers Outline the operation of three phase is an official topic in Module 2 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding semi controlled rectifiers Outline the operation of three phase using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding semi controlled rectifiers outline the operation of three phase should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding semi controlled rectifiers Outline the operation of three phase means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഘട്ടം 2 Understanding semi controlled rectifiers  
Outline the operation of three phase ഓട്ടം ഓട്ടം definition/meaning  
ഓട്ടം ഓട്ടം. ഓട്ടം ഓട്ടം ഓട്ടം, steps, application ഓട്ടം ഓട്ടം  
ഓട്ടം. Technical terms English-ഘട്ടം ഓട്ടം ഓട്ടം.

**3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers - Terms and definitions - controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers ( listing only ) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only)**

3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers - Terms and definitions - controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers ( listing only ) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only) is an official topic in Module 2 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers - Terms and definitions - controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers ( listing only ) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only) using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 3 understanding controlled rectifiers series test 1 1 single phase half wave controlled rectifiers – terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -types of controlled rectifiers ( listing only ) single phase half wave controlled rectifiers-circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode) single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) three phase bridge converter with resistive load (basic diagram with concept only) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding controlled rectifiers Series Test 1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers ( listing only ) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only) means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- 3 Understanding controlled rectifiers Series Test 1 1 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers ( listing only ) Single phase half wave controlled rectifiers-



circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only)   
 definition/meaning . . . , steps, application . . . . Technical terms English-   
 .

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize the principle and operation of controlled rectifiers.

|               |                                               |   |
|---------------|-----------------------------------------------|---|
|               | Illustrate the operation of single phase half |   |
| M2.01         |                                               | 4 |
| Understanding | wave controlled rectifier                     |   |
|               | Explain the operation of single phase full    |   |
| M2.02         |                                               | 5 |
| Understanding | wave controlled rectifier                     |   |
|               | Summarize the operation of single phase       |   |
| M2.03         |                                               | 2 |
| Understanding | semi controlled rectifiers                    |   |
|               | Outline the operation of three phase          |   |
| M2.04         |                                               | 3 |
| Understanding | controlled rectifiers                         |   |
|               | SeriesTest1                                   | 1 |

Contents:  
 Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers ( listing only )  
 Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode)  
 Single phase full wave controlled rectifiers-circuit diagram, waveforms

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

### MODULE 3

## Summarize the principle and operation of DC and AC converters.

This module is presented from the official syllabus scope for Power Electronics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Summarize the principle and operation of DC and AC converters.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## **Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost**

Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost is an official topic in Module 3 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, summarize the operation of dc choppers 3 understanding illustrate the classification of dc chopper 5 understanding m3.02 explain the operation of buck and boost should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### **Topic study guide**

| Aspect              | What to prepare                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-  
Summarize the operation of DC choppers 3 Understanding  
Illustrate the classification of DC chopper 5 Understanding M3.02  
Explain the operation of buck and boost   
 definition/meaning  
 , steps, application   
 . Technical terms English-  
 .

## 2 Understanding converters

2 Understanding converters is an official topic in Module 3 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding converters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding converters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding converters means in the official course context              |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-2 Understanding converters ഓരോന്നിനും  
definition/meaning നൽകണം. ഓരോന്നിന്റെ ഓരോന്നും, steps,  
application നൽകണം. Technical terms English-ൽ നൽകണം.  
ഓരോന്നിനും.

**Summarize the operation of Cycloconverters 2 Understanding DC Chopper-  
basic principle of operation with waveforms- equation of output voltage -  
simple problems- control strategies- constant frequency control - variable  
frequency control Types of choppers -concept of quadrant operations- circuit  
diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-  
twoquadrant(TypeC,TypeD)- fourquadrant ( Type E )-Applications of chopper  
Buck and boost converters-(operation with diagram only) Cycloconverters -  
types -single phase to single phase step up cycloconverter (midpoint type)**

Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)- twoquadrant(TypeC,TypeD)- fourquadrant ( Type E )-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters -types - single phase to single phase step up cycloconverter (midpoint type) is an official topic in Module 3 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)- twoquadrant(TypeC,TypeD)- fourquadrant ( Type E )-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters -types -single phase to single phase step up cycloconverter (midpoint type) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, summarize the operation of cycloconverters 2 understanding dc chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(typea,typeb)-twoquadrant(typec,typed)-fourquadrant ( type e )-applications of chopper buck and boost converters-(operation with diagram only) cycloconverters -types -single phase to single phase step up cycloconverter (midpoint type) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant ( Type E )-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters -types -single phase to single phase step up cycloconverter (midpoint type) means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant ( Type E )-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters -types -single phase to single phase step up cycloconverter (midpoint type)



definition/meaning. . . . ., steps, application . . . . . Technical terms English- . . . . .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize the principle and operation of DC and AC converters.

M3.01 Summarize the operation of DC choppers 3  
Understanding

Illustrate the classification of DC chopper 5  
Understanding

M3.02 Explain the operation of buck and boost 2  
M3.03 Understanding  
converters

M3.04 Summarize the operation of Cycloconverters 2  
Understanding

Contents:

DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control

Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB) -twoquadrant(TypeC,TypeD) - fourquadrant ( Type E )-Applications of chopper

Buck and boost converters-(operation with diagram only)

Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type)

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 4

# Summarize various inverters and electric drives

This module is presented from the official syllabus scope for Power Electronics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Summarize various inverters and electric drives
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## Explain the operation of basic inverters

Explain the operation of basic inverters is an official topic in Module 4 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the operation of basic inverters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the operation of basic inverters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                    |
|---------------------|------------------------------------------------------------------------------------|
| Definition/identity | What Explain the operation of basic inverters means in the official course context |
| Study lens          | Principle → components or stages → result → application                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Explain the operation of basic inverters  
പ്രവർത്തനം definition/meaning പ്രവർത്തനം. പ്രവർത്തനം പ്രവർത്തനം, steps, application പ്രവർത്തനം പ്രവർത്തനം. Technical terms English- പ്രവർത്തനം പ്രവർത്തനം.

## Illustrate the operation of bridge inverters

Illustrate the operation of bridge inverters is an official topic in Module 4 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Illustrate the operation of bridge inverters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate the operation of bridge inverters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                        |
|---------------------|----------------------------------------------------------------------------------------|
| Definition/identity | What Illustrate the operation of bridge inverters means in the official course context |
| Study lens          | Principle → components or stages → result → application                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-ഈ ഉപകരണങ്ങളുടെ പ്രവർത്തനം വിശദീകരിക്കുക  
ഈ ഉപകരണങ്ങളുടെ പ്രവർത്തനം വിശദീകരിക്കുക. ഈ ഉപകരണങ്ങളുടെ പ്രവർത്തനം,  
steps, application ഈ ഉപകരണങ്ങളുടെ പ്രവർത്തനം. Technical terms English-ഈ ഉപകരണങ്ങളുടെ  
പ്രവർത്തനം വിശദീകരിക്കുക.

## Explain different voltage control methods

Explain different voltage control methods is an official topic in Module 4 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain different voltage control methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain different voltage control methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                     |
|---------------------|-------------------------------------------------------------------------------------|
| Definition/identity | What Explain different voltage control methods means in the official course context |
| Study lens          | Principle → components or stages → result → application                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Explain different voltage control methods  
വോൾട്ടേജ് കൺട്രോൾ മെതോഡുകൾ വിശദീകരിക്കുക. definition/meaning, steps, application, technical terms English- Malayalam  
English- Malayalam.



**Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublishersporelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics <https://nptel.ac.in/courses/108102145><https://nptel.ac.in/courses/1081011038><https://nptel.ac.in/courses/108101126><https://nptel.ac.in/courses/108108077> 2 [www.electrical4u.com/electrical-engineering-articles/power-electronics](http://www.electrical4u.com/electrical-engineering-articles/power-electronics) 3 [www.swayam.gov.in](http://www.swayam.gov.in) 4 [www.youtube.com](http://www.youtube.com) 5 [wikipedia.org](http://wikipedia.org) 6. IEEE papers/journals on power electronics transactions**

Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types- advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,

(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136)  
 Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005  
 V.R.Moorthyoxfordpublisherspoverelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics  
<https://nptel.ac.in/courses/108102145><https://nptel.ac.in/courses/108101126><https://nptel.ac.in/courses/108108077>  
 1 2 [www.electrical4u.com/electrical-engineering-articles/power-electronics](http://www.electrical4u.com/electrical-engineering-articles/power-electronics) 3  
[www.swayam.gov.in](http://www.swayam.gov.in) 4 [www.youtube.com](http://www.youtube.com) 5 [wikipedia.org](http://wikipedia.org) 6. IEEE papers/journals on power electronics transactions is an official topic in Module 4 of Power Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types- advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136)  
 Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3

9788185749532) SenP.C.,Power Electronics,McGraw-

HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005

V.R.Moorthyoxfordpublishersporelectronicsdevicecircuitsandindustrial R5

applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical

Engineering>>Power Electronics

<https://nptel.ac.in/courses/108102145><https://nptel.ac.in/courses/108101126><https://nptel.ac.in/courses/108108077>

2 [www.electrical4u.com/electrical-engineering-articles/power-electronics](http://www.electrical4u.com/electrical-engineering-articles/power-electronics) 3

[www.swayam.gov.in](http://www.swayam.gov.in) 4 [www.youtube.com](http://www.youtube.com) 5 [wikipedia.org](http://wikipedia.org) 6. IEEE papers/journals

on power electronics transactions using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, classify various electric drives. 2 understanding inverter- requirements of a practical inverter- classification- circuit diagram and working - series inverter- parallel inverter- concept of csi and vsi bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for rl load). three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for r loads only. voltage control methods - principle of pulse width modulation- different voltage control methods-single pwm - sinusoidal pwm electric drives-introduction-types-advantages-basic block diagram text/reference: t/r book title/author p.s.bimbhra, power electronics, khanna publications,co.,new delhi(isbn: t1 9788174092151). m.d.singhand khanchandani k.b, power electronics.secondedition, mcgraw t2 hill publishing co. ltd, new delhi,2008, (isbn:9780070583894) muhammedhrashid, power electronics circuits devices and applications, r1 pearson education india,noida,2014.(isbn: 9788131702468) jainalok, power electronics and its applications, penram international r2 publishing,mumbai,2006 (isbn:9788187972136) dr.brgupta,singhal,powerelectronicsskkataria&sons,2009(isbn: r3 9788185749532) senp.c.,power electronics,mcgraw-hillpublishingcompanylimited,new delhi. r4 isbn:9780074624005

v.r.moorthyoxfordpublishersporelectronicsdevicecircuitsandindustrial r5 applications online resources sl.no website link nptel>>courses>>electrical engineering>>power electronics <https://nptel.ac.in/courses/108102145><https://nptel.ac.in/courses/108101126><https://nptel.ac.in/courses/108108077> 2 [www.electrical4u.com/electrical-engineering-articles/power-electronics](http://www.electrical4u.com/electrical-engineering-articles/power-electronics) 3 [www.swayam.gov.in](http://www.swayam.gov.in) 4 [www.youtube.com](http://www.youtube.com) 5 [wikipedia.org](http://wikipedia.org) 6. ieee papers/journals on power electronics transactions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

## Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | <p>What Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&amp;Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspawarelectronicsdevicecircuitsandindustrial R5 applications Online Resources SI.No Website Link NPTEL&gt;&gt;Courses&gt;&gt;Electrical Engineering&gt;&gt;Power Electronics</p> <p><a href="https://nptel.ac.in/courses/108102145">https://nptel.ac.in/courses/108102145</a><a href="https://nptel.ac.in/courses/108101126">https://nptel.ac.in/courses/108101126</a><a href="https://nptel.ac.in/courses/108108077">https://nptel.ac.in/courses/108108077</a> 2 <a href="https://www.electrical4u.com/electrical-engineering-articles/power-electronics">www.electrical4u.com/electrical-engineering-articles/power-electronics</a> 3 <a href="http://www.swayam.gov.in">www.swayam.gov.in</a> 4 <a href="https://www.youtube.com">www.youtube.com</a> 5 <a href="https://www.wikipedia.org">wikipedia.org</a> 6. IEEE papers/journals on power electronics transactions means in the official course context</p> |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid,

Power Electronics Circuits Devices and Applications, R1 Pearson Education India, Noida, 2014. (ISBN: 9788131702468) Jain Alok, Power Electronics and its Applications, Penram International R2 Publishing, Mumbai, 2006 (ISBN: 9788187972136)

Dr. B. R. Gupta, Singhal, Power Electronics K. Kataria & Sons, 2009 (ISBN: R3 9788185749532) Sen P. C., Power Electronics, McGraw-

Hill Publishing Company Limited, New Delhi. R4 ISBN: 9780074624005

V. R. Moorthy Oxford Publishers Power Electronics Device Circuits and Industrial R5 applications Online Resources Sl. No Website Link NPTEL >> Courses >> Electrical Engineering >> Power Electronics

<https://nptel.ac.in/courses/108102145> <https://nptel.ac.in/courses/108101103> <https://nptel.ac.in/courses/108101126> <https://nptel.ac.in/courses/108108077> 2

[www.electrical4u.com/electrical-engineering-articles/power-electronics](http://www.electrical4u.com/electrical-engineering-articles/power-electronics) 3

[www.swayam.gov.in](http://www.swayam.gov.in) 4 [www.youtube.com](http://www.youtube.com) 5 [wikipedia.org](http://wikipedia.org) 6. IEEE papers/journals on power electronics transactions

परिभाषा/अर्थ। परिभाषा परिभाषा परिभाषा, steps, application परिभाषा परिभाषा

परिभाषा. Technical terms English-हिंदी परिभाषा परिभाषा परिभाषा.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize various inverters and electric drives

|               |                                              |   |
|---------------|----------------------------------------------|---|
| M4.01         | Explain the operation of basic inverters     | 5 |
| Understanding |                                              |   |
| M4.02         | Illustrate the operation of bridge inverters | 7 |
| Understanding |                                              |   |
| M4.03         | Explain different voltage control methods    | 2 |
| Understanding |                                              |   |
| M4.04         | Classify various electric drives.            | 2 |
| Understanding |                                              |   |
|               | Series Test II                               | 1 |

Contents:

Inverter- requirements of a practical inverter- Classification- circuit diagram and working -

series inverter- parallel inverter- concept of CSI and VSI

Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three

phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only.

Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## RAPID REFERENCE

## Concept and formula bank

### Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

## RAPID REVISION

### Diagram and source-transcript library

#### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

## OFFICIAL SELF-LEARNING

# Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

## EXAMINATION PREPARATION

### Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

## ACTIVE RECALL

### Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

#### Module 1

##### Define or explain devices.. (3 marks)

**Answer:** Begin with the standard definition or identity of *devices*.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.



**Define or explain Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain the operation of SCR 3 Understanding Classify and explain turn on and turn off*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding methods of SCR.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding methods of SCR..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices -Symbols ( MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types - structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC-schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current-latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering-dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only). (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize the idea of datasheets of SCR 1 Understanding Power Electronic Devices- List various devices - Symbols ( MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list*

different types -structure and working of N-channel enhancement type-IGBT-- structure-working principle - applications-UJT- schematic diagram and working- DIAC & TRIAC- schematic diagram- working- V/I characteristics. Silicon Controlled Rectifier (SCR)- structure-V-I characteristics- definitions-holding current- latching current SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering- resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods-basic concepts-Thyristor protection circuits-snubber circuits. Specifications of SCR- List parameters (Datasheet interpretation only). State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 2

**Define or explain 4 Understanding wave controlled rectifier Explain the operation of single phase full. (3 marks)**

**Answer:** Begin with the standard definition or identity of 4 *Understanding wave controlled rectifier Explain the operation of single phase full*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 5 Understanding wave controlled rectifier Summarize the operation of single phase. (3 marks)**

**Answer:** Begin with the standard definition or identity of 5 *Understanding wave controlled rectifier Summarize the operation of single phase*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 2 Understanding semi controlled rectifiers Outline the operation of three phase. (3 marks)**

**Answer:** Begin with the standard definition or identity of 2 *Understanding semi controlled rectifiers Outline the operation of three phase*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 3 Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers - Terms and definitions - controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers ( listing only ) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only). (3 marks)**

**Answer:** Begin with the standard definition or identity of 3 *Understanding controlled rectifiers SeriesTest1 1 Single phase half wave controlled rectifiers - Terms and definitions - controlled rectifiers- firing angle-conduction angle-freewheeling diode - Types of controlled rectifiers ( listing only ) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation( R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers-circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only)*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 3

**Define or explain Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 2 Understanding converters. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding converters*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms-equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers- singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant ( Type E )-Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters -types -single phase to single phase step up cycloconverter (midpoint type). (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize the operation of Cycloconverters 2 Understanding DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers- singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)- fourquadrant ( Type E )- Applications of chopper Buck and boost converters-(operation with diagram only) Cycloconverters -types -single phase to single phase step up cycloconverter (midpoint type). State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

**Malayalam support:** □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

## Module 4

**Define or explain Explain the operation of basic inverters. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain the operation of basic inverters*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain Illustrate the operation of bridge inverters. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Illustrate the operation of bridge inverters*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain Explain different voltage control methods. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain different voltage control methods*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram**

**Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894)**

**MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014.(ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136)**

**Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005**

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**1 1038<https://nptel.ac.in/courses/108101126> 2 [www.electrical4u.com/electrical-engineering-articles/power-electronics](http://www.electrical4u.com/electrical-engineering-articles/power-electronics)**

**3 [www.swayam.gov.in](http://www.swayam.gov.in) 4 [www.youtube.com](http://www.youtube.com) 5 [wikipedia.org](http://wikipedia.org) 6. IEEE**

**papers/journals on power electronics transactions. (3 marks)**

**Answer:** Begin with the standard definition or identity of Classify various electric drives. 2 Understanding Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only. Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM Electric drives-introduction-types-advantages-basic block diagram Text/Reference: T/R Book Title/Author P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: T1 9788174092151). M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw T2 Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894) MuhammedHRashid, Power Electronics Circuits Devices and Applications, R1 Pearson Education India,Noida,2014. (ISBN: 9788131702468) JainAlok, Power Electronics and its Applications, Penram International R2 Publishing,Mumbai,2006 (ISBN:9788187972136) Dr.BRGupta,Singhal,PowerElectronicsKKataria&Sons,2009(ISBN: R3 9788185749532) SenP.C.,Power Electronics,McGraw-HillPublishingCompanyLimited,New Delhi. R4 ISBN:9780074624005 V.R.Moorthyoxfordpublisherspoverelectronicsdevicecircuitsandindustrial R5 applications Online Resources Sl.No Website Link NPTEL>>Courses>>Electrical Engineering>>Power Electronics <https://nptel.ac.in/courses/108102145><https://nptel.ac.in/courses/108101126><https://nptel.ac.in/courses/108108077> 1 2 3 [www.electrical4u.com/electrical-engineering-articles/power-electronics](http://www.electrical4u.com/electrical-engineering-articles/power-electronics) 4 [www.swayam.gov.in](http://www.swayam.gov.in) 5 [www.youtube.com](http://www.youtube.com) 6. IEEE papers/journals on power electronics transactions. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and



marks.

## Module 1

1-mark definition: State the meaning of **devices**..

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **4 Understanding wave controlled rectifier Explain the operation of single phase full.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **Summarize the operation of DC choppers 3 Understanding Illustrate the classification of DC chopper 5 Understanding M3.02 Explain the operation of buck and boost.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **Explain the operation of basic inverters.**

3-mark explanation: Explain one principle, process or comparison from this module.



7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

# Exam checklist

## Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

## Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

## REFERENCE VOCABULARY

# Glossary

### Study glossary

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

### References and web resources

#### References listed in the official PDF

References are included in the official source PDF.

#### Web resources listed in the official PDF

- Sl.No Website Link
- NPTEL>>Courses>>Electrical Engineering>>Power Electronics
- <https://nptel.ac.in/courses/108102145><https://nptel.ac.in/courses/108102145>
- <https://nptel.ac.in/courses/108101126><https://nptel.ac.in/courses/108101126>
- IEEE papers/journals on power electronics transactions

## IN-PAGE SEARCH

### Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 4541. Verify institutional updates against the current official curriculum.